

How Do Intermediaries Shape Labor Market Efficiency?

An Equilibrium Model and Experimental Evidence

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How Do Intermediaries Shape Labor Market Efficiency?

- Workers and firms often find each other through intermediaries
 - Job posting boards, university career fairs, union hiring halls, labor recruiters for seasonal migrant work, hiring agencies, talent agencies...
- We focus on three features present in many labor markets:
 1. Intermediaries compete with each other and have market power
 2. They can steer matches toward specific individuals
 3. They may charge participants for preferential treatment in matching.

This paper

- A large-scale experiment in partnership with an **online job platform**
- An equilibrium search and matching model with intermediaries
- Counterfactual simulations and discussion
 - What are the welfare effects of banning “premium tiers”?

Literature and contribution

- Labor literature on intermediaries and the internet:
 - AUTOR (2001); KUHN AND SKUTERUD (2004); AUTOR (2008); STEVENSON (2008); AUTOR AND HOUSEMAN (2010); BHULLER, FERRARO, KOSTØL AND VIGTEL (2023); FARBOODI, JAROSCH AND SHIMER (2023); SHIMER AND WU (2023); FARBOODI, JAROSCH, MENZIO AND WIRIADINATA (2025); CRÉPON ET AL. (2013); PALLAIS (2014); CHERON AND DECREUSE (2016); KAISER (2025)
- Industrial organization of intermediated markets:
 - Models with single platforms: DINERSTEIN, EINAV, LEVIN AND SUNDARESAN (2018), HORTON, JOHARI AND KIRCHER (2024), LEE AND MUSOLFF (2025), and many others
 - Competition between platforms: ROCHET AND TIROLE (2003); CAILLAUD AND JULLIEN (2003); HAGIU AND JULLIEN (2014); ATKESON, EISELDT AND WEILL (2015); ROSAIA (2025)
- Contribution: intermediaries with market power who choose entry, steering, and monetization strategies

Literature and contribution

- Experiments in online labor markets:
 - HORTON (2017), BARACH AND HORTON (2021), HORTON (2025), FILIPPAS, HORTON, PARASURAMA AND URRACA (2025*b*), HORTON, BARACH AND GOLDEN (2020), FILIPPAS, HORTON, PARASURAMA AND URRACA (2025*a*)
- Advertising experiments, saliency, and signaling:
 - NARAYANAN AND KALYANAM (2015); URSU (2018); ABHISHEK, JERATH AND SHARMA (2025); BLAKE, NOSKO AND TADELIS (2015); SIMONOV, NOSKO AND RAO (2018); SAHNI AND NAIR (2019); SAHNI AND ZHANG (2024); MOSHARY (2025)
- We provide a framework to interpret and extrapolate experimental results.

Online Job Platforms

- Websites where employers hire temporary, remote labor (*freelancers*)
- Our platform features many types of jobs:
 - On the low end: Copy editors, translators, data entry
 - On the high end: programmers with \$1M+ in cumulative earnings

Platform revenue sources

- Match fees ($\sim 90\%$ of revenues in our platform)
 - Proportional fee on worker earnings
 - Feasible because the platform intermediates payments, offers conflict resolution/monitoring tools, and workers benefit from building a history within the platform
- Premium membership fees ($\sim 10\%$)
 - Monthly payments in return for promised higher match rates
- Participation fees are rare. [▶ Go](#)

Better sorting or rent extraction?

- Premium memberships may plausibly redirect employers to the right workers
 - Easy and cheap to create worker accounts; many are never matched (KÄSSI, LEHDONVIRTA AND STEPHANY, 2021)
 - Paying for premium memberships: costly signal of readiness for matches
- But they may also be tools for rent extraction
 - Implemented by restricting access to zero-marginal-cost resources (messaging, job applications, access to information, etc)
 - Consistent with a “damaged goods” story (DENECKERE AND MCAFEE, 1996)

The Experiment

- Our platform already had premium memberships for workers
- The experiment aimed to answer the following question: *What happens to hiring rates if the platform increases the degree of **match-steering** toward premium-tier workers?*

The Experiment

- Period: May-July 2019
- Randomization at the *employer* level, 50/50 split [▶ Balance](#)
- Treated employers see ads at the top of the page when searching for workers
 - Advertised workers: those with premium memberships [▶ Figure](#)
- Non-treated employers never see any ads for workers
- Important quantities:
 - 465K employers, 275K execute at least one search, 169K hired someone
 - 1.7mi workers, 159K were hired
 - [▶ Data](#)

Experimental results—employer clicking behavior

	All workers	Sometimes premium	Never premium
<i>Panel A: dependent variable is any clicks on these workers</i>			
Treatment	-0.00413*** (0.00147)	0.0155*** (0.00128)	-0.0121*** (0.00133)
Constant	0.485*** (0.00104)	0.248*** (0.000897)	0.299*** (0.000951)
N	464,986	464,986	464,986

Experimental results—employer hiring decisions

	All workers	Sometimes premium	Never premium
<i>Panel C: dependent variable is any hire</i>			
Treatment	-0.00263** (0.00133)	0.0000115 (0.000941)	-0.00252*** (0.00102)
Constant	0.293*** (0.000945)	0.117*** (0.000666)	0.143*** (0.000728)
N	464,986	464,986	464,986

Experimental results—summary

- Treated employers are **0.9%** less likely to hire anyone, and **1.8%** less likely to hire a worker with a basic membership
- Preferred interpretation:
 - Employers have limited attention and ads make premium workers more **salient**
 - A premium worker “forced” into the consideration set is a worse match than the non-premium one it displaces.
- We can rule out some alternative interpretations [▶ Go](#)

Why do we need an equilibrium model?

1. Spillovers in the experimental design: treated and control employers are exposed to the same workers
2. Even if a marginal increase in steering reduces employment, some positive level of steering may still be socially optimal
3. Welfare and employment are not synonymous: search costs also matter
4. To predict the consequences of policy changes or other counterfactuals, we need to account for intermediaries' profit-maximizing choices—including monetization strategies—as they compete with one another

What's new relative textbook search and matching?

- **# 1:** Intermediaries (or platforms) as profit-maximizing agents, in addition to workers and firms
- **# 2:** A microfoundation for worker-firm matching inside platforms [▶ Go](#)
 - Platforms choose how often to steer matches toward premium-tier workers, $\gamma \in [0, 1]$
 - Steering can either increase or decrease vacancy-filling rates, depending on parameters that are recovered from the experimental results
- **# 3:** A richer formulation of worker and firm flows across possible states

Unemployed workers

Job-finding rate

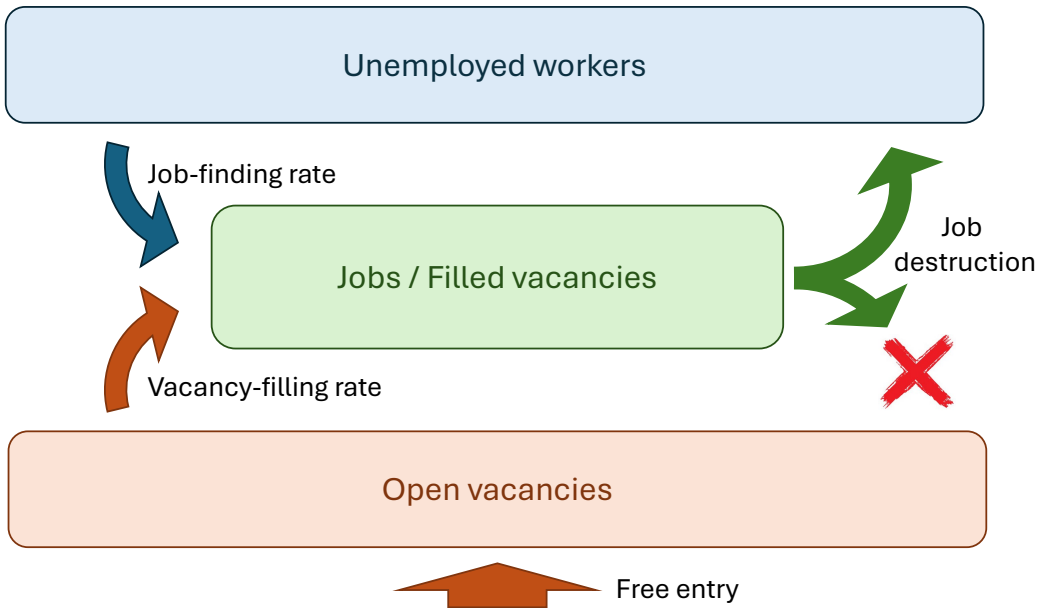
Jobs / Filled vacancies

Job
destruction

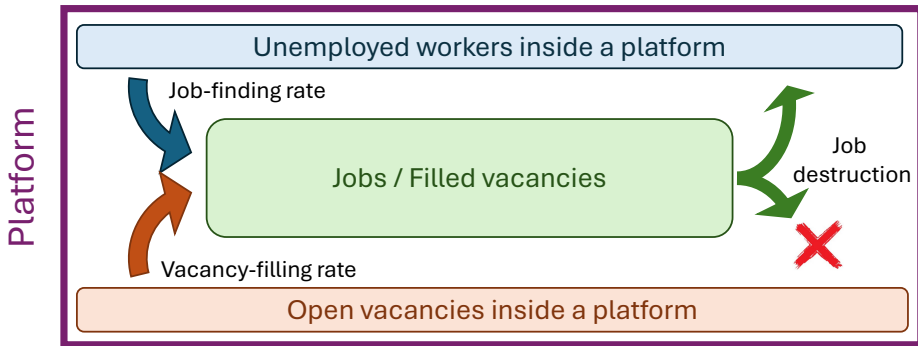
Vacancy-filling rate

Open vacancies

Free entry



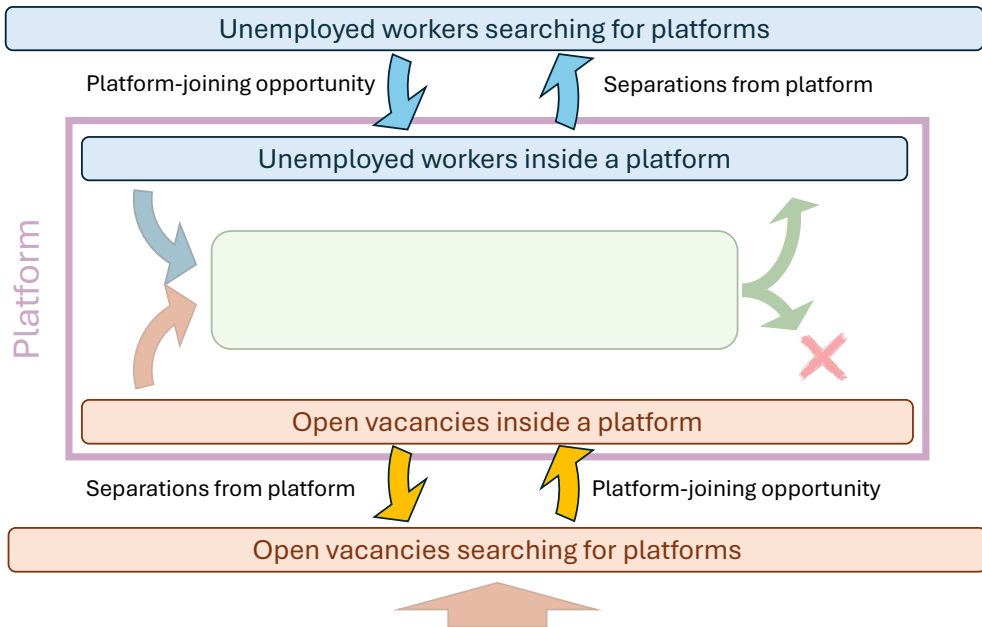
Unemployed workers searching for platforms

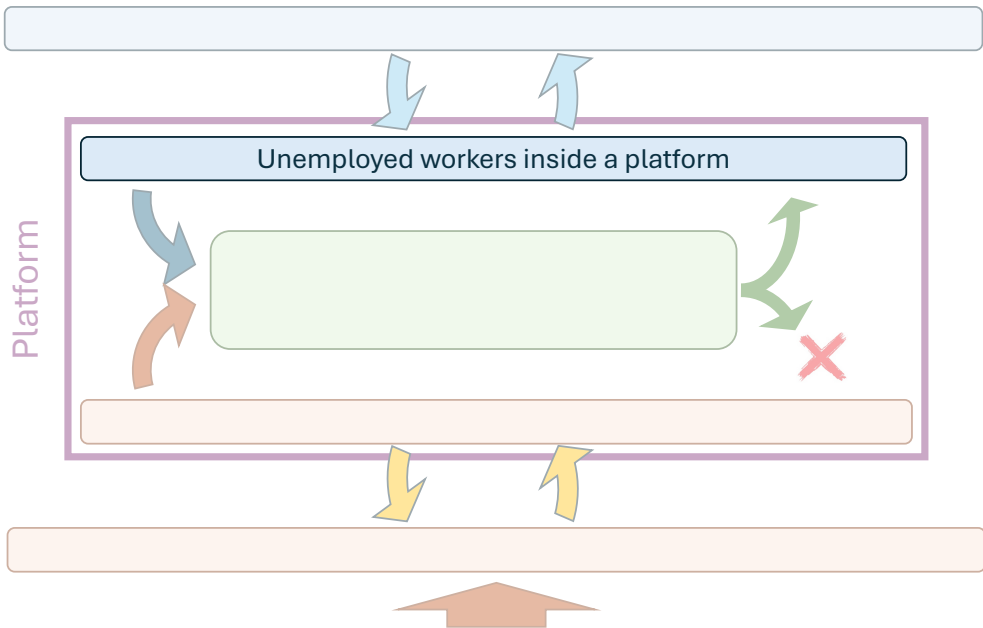


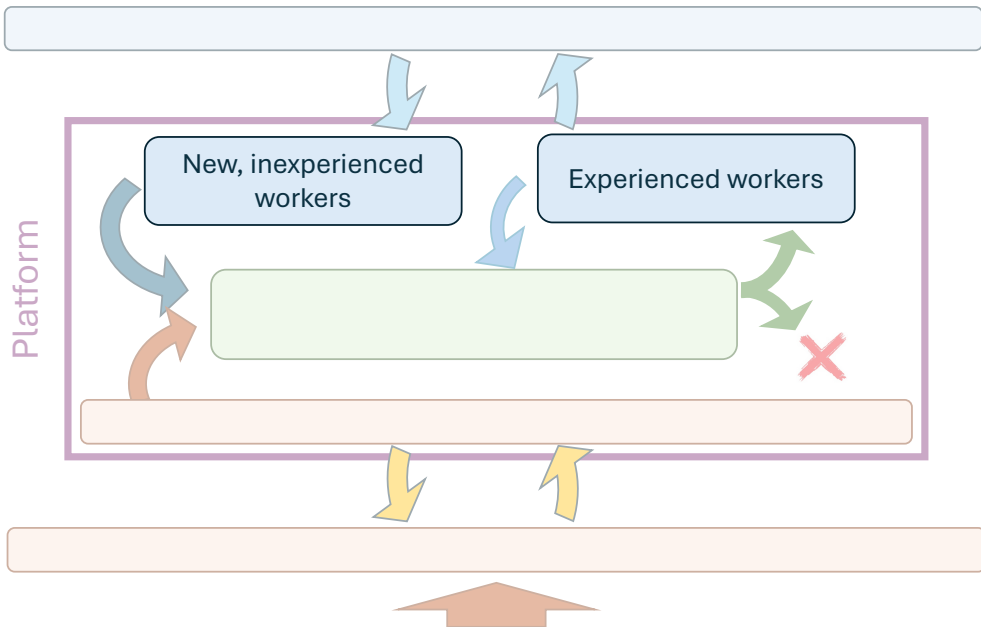
Open vacancies searching for platforms

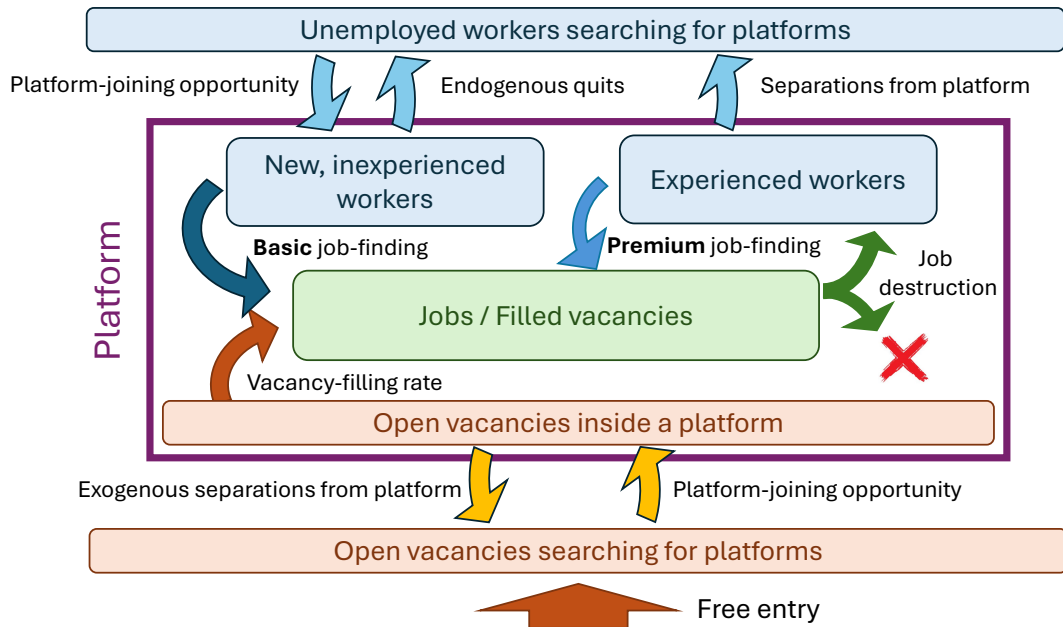


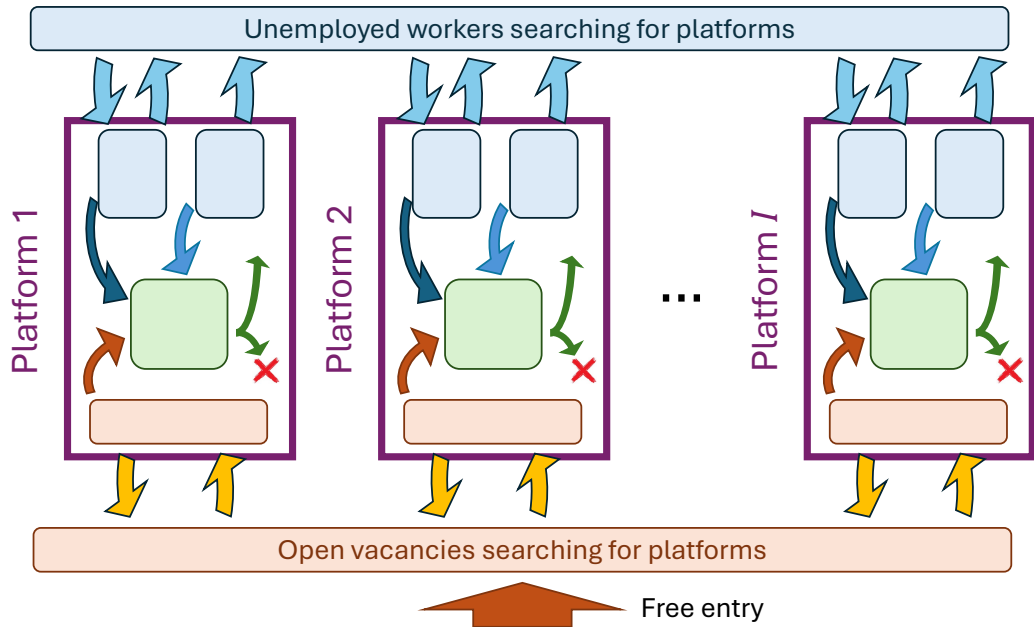
Free entry











The Problem of the Platform

$$\begin{aligned}\Pi(\text{Agg. vars}) = & \max_{\text{Fees, steering}} (\text{Match fees}) + (\text{Premium fees}) - (\text{Fixed flow cost}) \\ \text{s.t.} \quad & \text{Experienced worker chooses profit-maximizing tier.}\end{aligned}$$

- **Key insight:** two revenue sources are substitutes, not complements
 - Higher match fees make premium fees less appealing, and vice-versa
 - Model admits corner solutions—consistent with real platform behavior
 - Premium memberships more appealing when separations are less elastic than inflows

More on the model

- Imperfect information regarding job-finding [▶ Go](#)
- Value functions [▶ Go](#)
- Equilibrium conditions [▶ Go](#)
- **Extension 1:** Private information regarding suitability [▶ Go](#)
- **Extension 2:** Phantom accounts [▶ Go](#)

Calibration

- Our platform is one of many platforms in the model's symmetric, steady-state equilibrium
- Seven parameters are externally calibrated/normalized
- Eleven parameters calibrated by matching a corresponding set of eleven targets in the data
 - Targets include the experimental results on hiring probabilities
 - Experiment in the model: the focal platform increases the steering probability by $\Delta\gamma$, but only for vacancies assigned to the treatment group
 - [► Calibration details](#)

Premium Ban: platforms must set $\gamma = f_p = 0$

Outcome	Mkt. Equilibrium	Premium Prohibition
<i>Panel A: Aggregate metrics</i>		
Welfare gains (%)	-	4.2
Welfare gains relative to initial aggregate payroll (%)	-	3.2
Employment rate	0.100	0.103
Search costs as share of output	0.586	0.582
<i>Panel B: Market outcomes</i>		
Number of platforms	10.0	10.2
Match fees	0.150	0.163
Match distortion γ	0.046	-
Premium membership fee	0.34	-

- Welfare is output net of all search costs

Additional exercises

- How does the experiment inform the previous counterfactual? [▶ Go](#)
- Optimal platform entry [▶ Go](#)
- Premium memberships, search costs, and the Beveridge curve [▶ Go](#)
- Premium memberships and competition between intermediaries [▶ Go](#)
- Discussion: applicability to other labor markets [▶ Go](#)
 - Talent agents for actors/models/screenwriters
 - Seasonal migrant work in developing countries

Conclusion

- Experimental evidence on the impact of steering employers' attention in online labor markets
- An equilibrium model of labor market intermediaries that make choices on entry, match-steering, and monetization strategies
- “Premium tiers” seem to degrade market efficiency in the context we study, despite their potential to reduce congestion externalities
- Ideas for the future:
 - Use this framework in other settings
 - Study how competition and monetization interact with other platform choices, e.g., algorithmic recommendations (HORTON, 2017)

Thanks!

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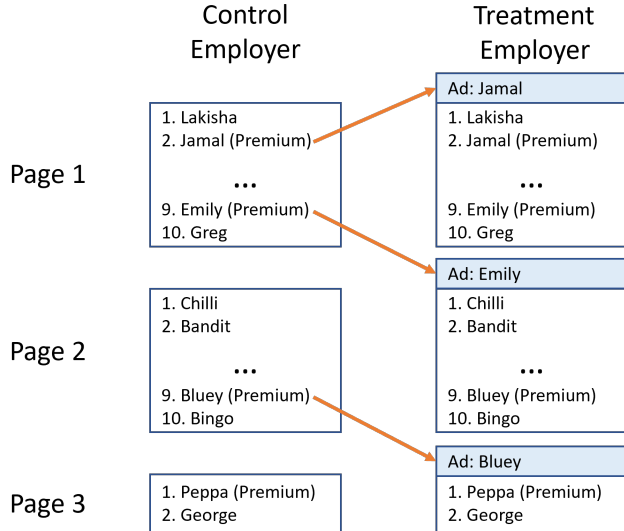
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Platform	Fees on Matches	Premium for Workers	Premium for Employers	Participation Fee
99designs	Yes	No	Yes	No
Care.com	No	Yes	Yes	Yes
Catalant	Yes	No	Yes	No
DesignCrowd	Yes	No	Yes	No
Fiverr	Yes	Yes	Yes	No
Freelancer	Yes	Yes	Yes	No
Guru	Yes	Yes	Yes	No
Handy	Yes	No	No	No
Hubstaff Talent [†]	No	No	No	No
Outsourcely	No	Yes	Yes	Yes
PeoplePerHour	Yes	Yes	No	No
Rover	Yes	Yes	No	No
TaskRabbit	Yes	No	No	No
Thumbtack	Yes ^{††}	No	No	No
Toptal	Yes	No	No	No
Truelancer	Yes	Yes	No	No
Upwork	Yes	Yes	Yes	No
WeGoLook	Yes	No	No	No
Wonolo	Yes	No	No	No
Workana	Yes	Yes	Yes	No

Randomization Checks

Variable	Treatment Mean	Control Mean	P-value
Number of Searches	1.2805	1.3051	0.0191**
Has Searched (1/0)	0.3044	0.3046	0.8627
Number of Invitations	1.8419	2.0377	0.0641*
Has Invited (1/0)	0.1303	0.1313	0.2811
Number of Hires	0.3749	0.3757	0.9157
Has Hired (1/0)	0.1337	0.1350	0.2173
Number of Posts	8.8312	8.8424	0.9207
Has Posted (1/0)	0.4202	0.4216	0.3596
Total Spent	4,915.86	4,882.39	0.8839
Has Spent (1/0)	0.3204	0.3209	0.6818
Experience at Entry	522.06	524.06	0.4321
US (1/0)	0.4351	0.4359	0.5754
Joint Test	—	—	0.3654



Data

- Detailed information on searches: what was shown on the employer's screen and what was clicked
- Invitations to apply for open vacancies and hires
- Time stamps
- Premium membership status (for a non-random subset of workers)
- Hourly wages (only for hourly contracts, about half of all hires)
- No data on ratings

Alternative interpretations

- Maybe treated employers are exposed to a different set of workers?
 - The experimental design aims to minimize this possibility
 - Pairwise regressions that include worker fixed effects yield similar results [▶ Go](#)
- Maybe advertising provides a **signal** to employers?
 - To explain our results, advertising would have to convey a negative signal
 - This is not consistent with click behavior
 - It is even less consistent with the finding that treated employers are more likely to invite premium workers to apply to their vacancies [▶ Go](#)
 - Our model allows for advertising to convey a signal to *the platform*, not to employers

	Hired	Clicked
Treatment _j	-0.000448*** (0.000149)	-0.00315*** (0.000575)
Treatment _j × Premium _{i,t}	0.000502* (0.000267)	0.00238** (0.000948)
N	415566	
r ²	0.143	0.120
Unique employers	223187	
Unique workers	22040	
Worker fixed effects	✓	✓

	All workers	Sometimes premium	Never premium
<i>Panel A: dependent variable is any invitation sent to these workers</i>			
Treatment	-0.00168 (0.00142)	0.00566*** (0.00116)	-0.00543*** (0.00124)
Constant	0.377*** (0.00101)	0.193*** (0.000819)	0.237*** (0.000883)
<i>Panel B: dep. var. is log(1 + number of invitations sent to these workers)</i>			
Treatment	-0.00742*** (0.00285)	0.00673*** (0.00146)	-0.00879*** (0.0173)
Constant	0.624*** (0.00203)	0.210*** (0.00103)	0.283*** (0.00124)
N	464986	464986	464986

Imperfect information regarding job-finding

- Platform-searching workers do not observe job-finding rates $h_{b,i}, h_{p,i}$ —instead, they use priors $\bar{h}_b, \bar{h}_p$
- These beliefs are consequential for platform choices, even though workers are correct in the symmetric equilibrium
- Workers learn true job-finding rates upon the first match
- In the paper: argument for why this assumption seems more consistent with the data, along with a robustness exercise

◀ Back

Value functions, 1/3: Firms

- Newly-created vacancy (platform search):

$$rV_0^e = -c + \lambda_0 \int_0^\infty \left[\frac{1}{I} \sum_i \max \left\{ V_i^e - \epsilon - V_0^e, 0 \right\} \right] \sigma_\epsilon e^{-\sigma_\epsilon \epsilon} d\epsilon,$$

- ϵ : idiosyncratic joining cost (Exponentially distributed).
- Open vacancy within intermediary (platform) $i \in \{1, \dots, I\}$:

$$rV_i^e = -c + q_i [J_i^e - V_i^e] + s_V [V_0^e - V_i^e],$$

s_V : exogenous separations.

- Filled job in platform i :

$$rJ_i^e = y - w_i + \delta [0 - J_i^e],$$

Value functions, 2/3: Inexperienced workers

- Worker searching for platforms:

$$rV_0^w = \lambda_0 \int_0^\infty \left[\frac{1}{I} \sum_i \max \left\{ \tilde{V}_i^{iw}(0) - V_0^w - \epsilon, 0 \right\} \right] \sigma_\epsilon e^{-\sigma_\epsilon \epsilon} d\epsilon,$$

- Perceived* value function for inexperienced worker:

$$r\tilde{V}_i^{iw}(t) = \bar{h}_b \left[\tilde{J}_i^w - \tilde{V}_i^{iw}(t) \right] \Pr(x = 1|t, \bar{h}_b) + \frac{d\tilde{V}_i^{iw}(t)}{dt}.$$

- t is time since joining the platform
- $\bar{h}_b$: perceived job-finding rate for basic-membership workers

Value functions, 3/3: Experienced workers

- Employed workers:

$$rJ_i^w = (1 - f_{m,i})w + \delta \left[\max \{ V_i^b, V_i^p \} - J_i^w \right]$$

- $f_{m,i}$: match fees

- Unemployed, but experienced workers, by membership tier:

$$rV_i^b = h_{b,i} \left[J_i^w - V_i^b \right] + s_U \int_0^\infty \max \{ -\nu, V_0^w - V_i^b \} \sigma_\nu e^{-\sigma_\nu \nu} d\nu$$

$$rV_i^p = -f_{p,i} + h_{p,i} \left[J_i^w - V_i^p \right] + s_U \int_0^\infty \max \{ -\nu, V_0^w - V_i^p \} \sigma_\nu e^{-\sigma_\nu \nu} d\nu,$$

- $f_{p,i}$: premium membership fee
- ν : idiosyncratic disutility shock (avoidable through separation)

Equilibrium conditions

- Wages set by Nash bargaining
- Inflows to and outflows from all “pools” in the model are equalized
- Inexperienced workers leave platforms after the optimal stopping time

$$T_i = \max \left\{ 0, \frac{1}{\bar{h}_b} \left[\ln \left(\frac{\bar{x}}{1 - \bar{x}} \right) + \ln \left(1 + \frac{\bar{h}_b}{r} \frac{\tilde{J}_i^w - V_0^w}{V_0^w} \right) \right] \right\}. \quad (1)$$

- Platform choices $(f_{m,i}, f_{p,i}, \gamma_i)$ maximize steady-state flow profits given aggregate variables.
- Free entry of platforms and firms; market clearing for workers

Extension #1: Private information on suitability

- Upon joining a platform, three things may happen:
 - With prob. θ , the worker is suitable ($x = 1$) and knows it, but cannot convey that information to the platform.
 - With prob. $(1 - \theta)\bar{x}$, the worker is suitable, but has no information about it.
 - With prob. $(1 - \theta)(1 - \bar{x})$, the worker is unsuitable and does not know it.
- Premium memberships allow the θ -shock workers to signal to the platform that they deserve preferential treatment in the matching process

◀ Back

Extension #2: Phantom accounts

- Workers who separate from the platform may leave inactive (phantom) accounts (CHERON AND DECREUSE, 2016)
- Platform can detect and eliminate phantom accounts at rate $s_{\dagger}$
- Phantom accounts belong to the basic tier; increase odds that premium memberships are efficient
- Can also be interpreted as a reduced-form for **multi-homing**

Calibration details

- Key idea: our platform is one of many identical platforms
- Several identical, segmented sub-markets for workers
 - About 10,000 workers in a segmented market
 - Examples: “Unity Developer,” “SEO Specialist,” “Startup Lawyer”
 - About 10 platforms serving each sub-market
- Employers in the data: all included in the experiment
- Workers in the data: all who are shown by the platform to those employers, either as search results or suggestions, within 60 days after assignment into the experiment

Externally calibrated parameters

Parameter	Value	Rationale
r	0.004	Yearly discount of 5%
δ	1.00	Av. emp. spell of one month (see Appendix)
N_w	10000	Normalization
y	100.0	Normalization
β	0.50	Symmetric bargaining
λ_0	150.00	Five arrivals per day while platform-searching
D	13500.00	Other values considered in additional exercises.

- Unit of time set to one month

Param.	Value	Estimation target	Data	Sim.
F	920.0	Number of platforms	<i>10.00</i>	10.00
s_U	0.923	Probability that an experienced worker does not change state in one month	<i>0.600</i>	0.600
s_V	0.311	Share of platform vacancies eventually filled	0.190	0.190
λ	2406.6	Average number of jobs found by workers over two months	0.054	0.054
c	5.505	Ratio of workers to firms in platform	3.549	3.549
σ_ϵ	0.017	Match fees as fraction of gross wage	0.150	0.150
σ_ν	0.235	Premium fees as fraction of platform revenue	0.100	0.100
$\bar{x}$	0.691	Share of premium-membership workers in platform	0.095	0.095
θ	0.152	Share of one-month workers choosing premium	0.122	0.122
$\Delta\gamma$	0.0180	Treatment effect: prob. hiring basic membership workers	-0.0176	-0.0176
$s_\dagger$	0.127	Treatment effect: prob. hiring any worker	-0.0090	-0.0090

How Does the Experiment Inform Counterfactuals?

- What if the experimental regression coefficients had been different?
 - Effect on probability of hiring **any worker**
 - Effect on probability of hiring a **basic-membership worker**
- New exercise:
 1. Pretend one, or both, coefficients are 10% smaller in magnitude
 2. Re-calibrate the model
 3. Re-do the “premium ban” counterfactual

How Does the Experiment Inform Counterfactuals?

Outcome	Baseline estimation	Both outcomes 10% lower	Any hiring 10% lower
<i>Panel A: Predicted gains from prohibition of premium memberships</i>			
Welfare gain (%)	4.21	4.22	3.76
<i>Panel B: Estimated parameters</i>			
$\bar{x}$	0.691	0.692	0.676
$s_{\dagger}$	0.127	0.127	0.119
$\Delta\gamma$	0.018	0.016	0.018
<i>Panel C: Composition of basic accounts in the platform</i>			
Suitable inexperienced workers	0.16	0.16	0.13
Unsuitable inexperienced workers	0.07	0.07	0.06
Phantoms	0.77	0.77	0.80

Optimal platform entry

- As a welfare benchmark, consider a **planner** that can choose the number of operating platforms and all platform choices
- Planner is subject to search frictions and constraints on revenue-raising strategies (e.g., no participation fees)
- Decentralized entry can be higher or lower than optimal entry:
 - Platforms do not fully internalize gains to workers and firms
 - Model features a **thin-market effect** which depends on D [▶ Go](#)
 - Baseline D is chosen such that decentralized entry is optimal; we consider alternative values for D as well

Outcome	$D = 8437.5$	$D = 13500.0$	$D = 21600.0$
<i>Panel A: Planner solution</i>			
Welfare gain (%)	4.9	4.2	4.8
Change in employment rate (%)	6.3	3.2	1.9
Change in gross output (%)	6.3	3.2	1.9
Change in search costs as share of output (p.p.)	-0.4	-0.4	-0.5
Number of platforms	12.8	10.0	7.9
Match fees	0.195	0.161	0.130
Match distortion γ	0.000	0.000	0.000
Premium membership fee	0.00	0.00	0.00
<i>Panel B: Premium membership prohibition policy</i>			
Welfare gain (%)	4.1	4.2	4.2
Change in employment rate (%)	3.1	3.4	3.5
Change in gross output (%)	3.1	3.4	3.5
Change in search costs as share of output (p.p.)	-1.4	-0.3	0.4
Number of platforms	10.1	10.2	10.2
Match fees	0.162	0.163	0.163

Thin market effect

- Recall from the model:

$$q_i = \frac{\lambda}{n_v} \left\{ \gamma_i \left[1 - e^{-n_{p,i}/D} \right] + (1 - \gamma_i) \left[1 - e^{-(n_{b,j} + n_{p,i})/D} \right] \phi_i \right\}.$$

- As $D, \lambda \rightarrow \infty$, but λ/D remains constant:

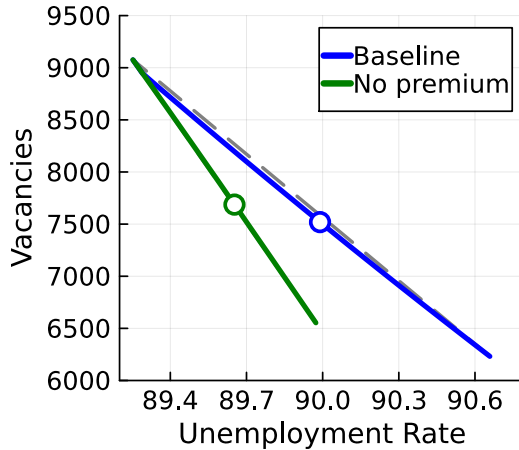
$$q_i = \frac{1}{n_v} \frac{\lambda}{D} \left[\gamma_i n_{p,i} + (1 - \gamma_i) \phi_i (n_{b,j} + n_{p,i}) \right].$$

- Constant-returns match technology! No reason to create multiple platforms...

Premium tiers, search costs, and the Beveridge curve

- We can trace the model-implied *vacancies* $\times$ *unemployment locus* by changing match productivity y
- In our setup: equivalent to shifting all search costs relative to output
- Two goals:
 - Is our model's macroeconomic behavior very different from standard search-and-matching models?
 - How do premium memberships interact with search costs?

Premium tiers, search costs, and the Beveridge curve



Platform Differentiation and Market Outcomes

- Counterfactual: reduce idiosyncratic platform-joining costs through a **10% increase in ρ_ϵ**
- In a model *without premium memberships*:
 - Worker and firm inflows into the platform become more elastic to perceived value $\Rightarrow$ match fees f_m fall
 - Profits per client fall $\Rightarrow$ larger platforms in equilibrium
- Welfare effects:
 - Direct positive effect through agent utilities
 - Additional effects (positive or negative) from changing # of platforms

Platform Differentiation and Market Outcomes

Outcome	$D = 8437.5$	$D = 13500.0$	$D = 21600.0$
<i>Panel A: Restricted model with premium memberships prohibited</i>			
Welfare gains from falling joining costs (%)		0.41	
Change in number of platforms I (%)		-1.95	
Change in match fees f_m (%)		-1.74	
<i>Panel B: Baseline model with premium memberships</i>			
Welfare gains from falling joining costs (%)			
Change in number of platforms I (%)			
Change in match fees f_m (%)			
Change in premium membership fees f_p (%)			

Platform Differentiation and Market Outcomes

Outcome	$D = 8437.5$	$D = 13500.0$	$D = 21600.0$
<i>Panel A: Restricted model with premium memberships prohibited</i>			
Welfare gains from falling joining costs (%)	0.08	0.41	0.65
Change in number of platforms I (%)	-1.99	-1.95	-1.97
Change in match fees f_m (%)	-1.53	-1.74	-1.93
<i>Panel B: Baseline model with premium memberships</i>			
Welfare gains from falling joining costs (%)			
Change in number of platforms I (%)			
Change in match fees f_m (%)			
Change in premium membership fees f_p (%)			

Platform Differentiation and Market Outcomes

Outcome	$D = 8437.5$	$D = 13500.0$	$D = 21600.0$
<i>Panel A: Restricted model with premium memberships prohibited</i>			
Welfare gains from falling joining costs (%)		0.41	
Change in number of platforms I (%)		-1.95	
Change in match fees f_m (%)		-1.74	
<i>Panel B: Baseline model with premium memberships</i>			
Welfare gains from falling joining costs (%)		-0.58	
Change in number of platforms I (%)		-2.38	
Change in match fees f_m (%)		-3.89	
Change in premium membership fees f_p (%)		23.98	

Platform Differentiation and Market Outcomes

Outcome	$D = 8437.5$	$D = 13500.0$	$D = 21600.0$
<i>Panel A: Restricted model with premium memberships prohibited</i>			
Welfare gains from falling joining costs (%)	0.08	0.41	0.65
Change in number of platforms I (%)	-1.99	-1.95	-1.97
Change in match fees f_m (%)	-1.53	-1.74	-1.93
<i>Panel B: Baseline model with premium memberships</i>			
Welfare gains from falling joining costs (%)	-0.80	-0.58	-0.40
Change in number of platforms I (%)	-2.22	-2.38	-2.48
Change in match fees f_m (%)	-3.37	-3.89	-4.26
Change in premium membership fees f_p (%)	21.94	23.98	25.20

Applicability to other labor markets

Our model is most applicable to markets where:

- Intermediaries are essential for match-making due to significant search frictions
- There are many intermediaries
- Workers and intermediaries form long-term relationships over multiple employment spells

Applicability to other labor markets

Talent agencies for actors, models, or screenwriters:

- Descriptive work emphasizes short employment spells and the importance of intermediaries (CHRISTOPHERSON AND STORPER, 1989; MENDER, 1999; BIELBY AND BIELBY, 1999)
- California regulations prevent “pay-to-play:” agencies may only charge percentage fees on successful hires
- Union agreements further restrict monetization strategies
- Despite that, reports of violations are common...

BACKSTAGE

Pay-to-Play Workshops Dealt Blow by L.A. City Attorney

BY PIYALI SYAM | LAST UPDATED: FEBRUARY 13, 2017

DEADLINE

SAG-AFTRA Puts Casting Websites On Notice For Charging Fees: “Performers Must Have A No-Cost Method To Submit Themselves”



By [Lynette Rice](#) 
January 20, 2024 9:19am

BACKSTAGE

Actor Alleges Bribery at N.Y. Casting Office

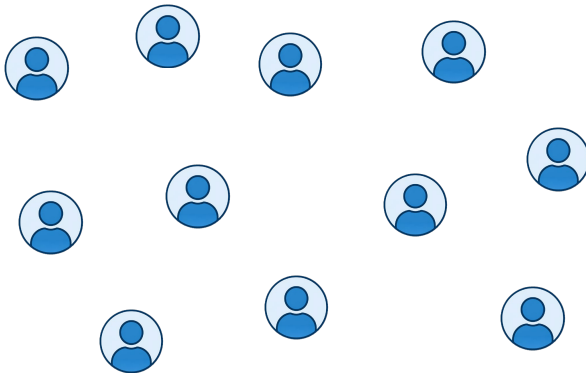
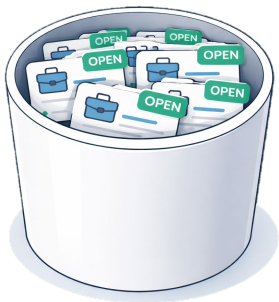
BY ANDREW SALOMON | LAST UPDATED: NOVEMBER 4, 2019

Applicability to other labor markets

Seasonal migrant work in developing countries:

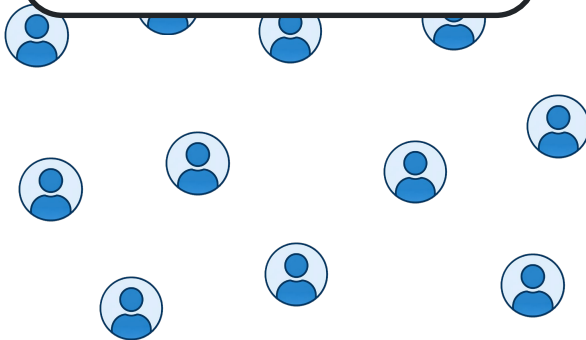
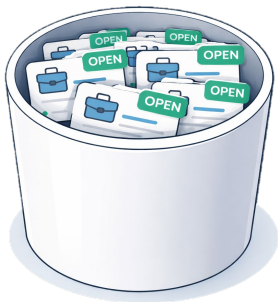
- An important source of income, but risky (BRYAN, CHOWDHURY AND MOBARAK, 2014)
- A government program facilitating seasonal migration from Tonga/Vanuatu to New Zealand had positive effects on households (GIBSON AND MCKENZIE, 2014)
 - The program placed restrictions on intermediaries' monetization strategies
- A program facilitating India→UAE seasonal migration had more ambiguous effects (NAIDU, NYARKO AND WANG, 2023)
 - Agent fees amount to 40% of annual household income in their sample.

At any point in time,
platform has open
vacancies and
unemployed workers.

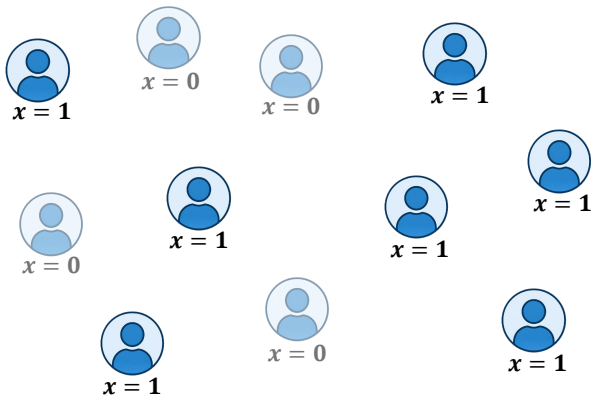




A particular worker-vacancy pair can be matched if two things are true: (i) the ***worker is suitable*** to the platform and (ii) the ***match distance*** is small.



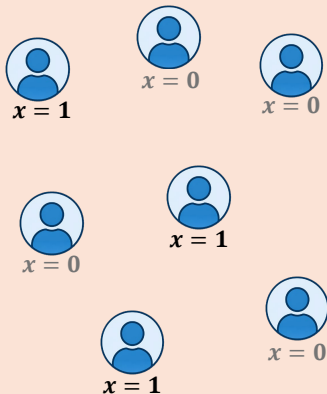
Worker suitability $x \in \{0,1\}$ is drawn when the worker joins the platform. If $x = 0$, the worker cannot be matched.



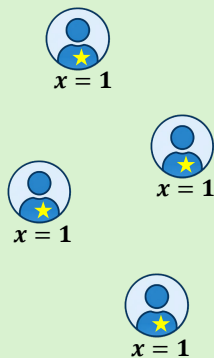
Suitability is not observed, but workers and platform infer that $x = 1$ upon the first successful match.



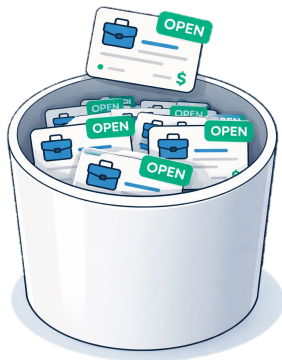
Inexperienced



Experienced

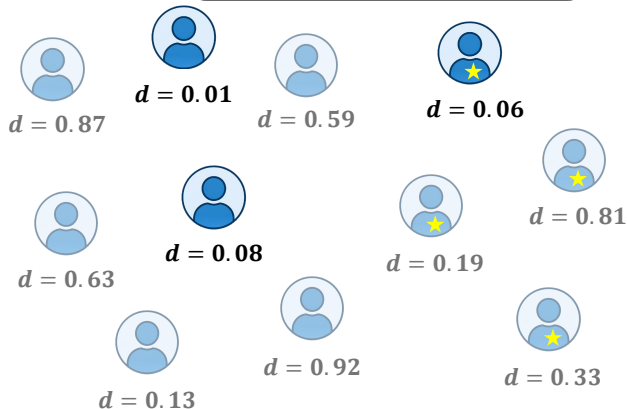


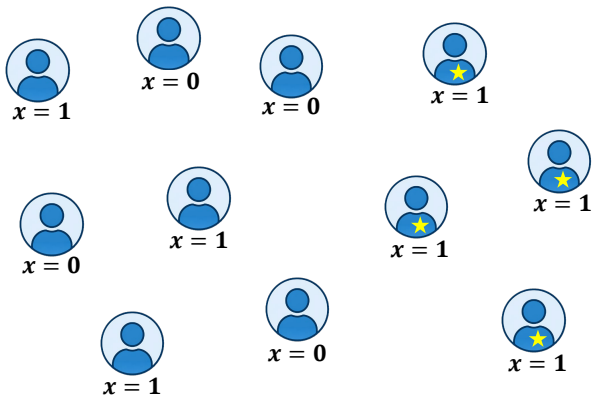
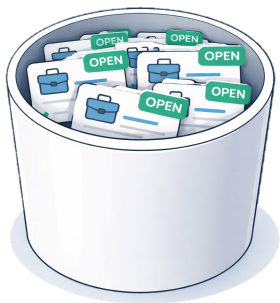
Match distance $d \sim U(0,1)$
is idiosyncratic at the
vacancy-worker-time level.



A match is viable only if
$$d < \frac{1}{D}$$

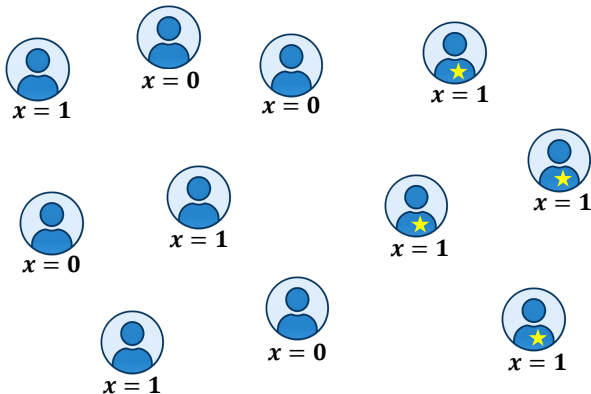
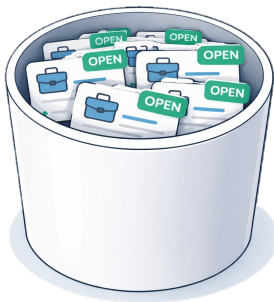
In this example, $D = 10$.



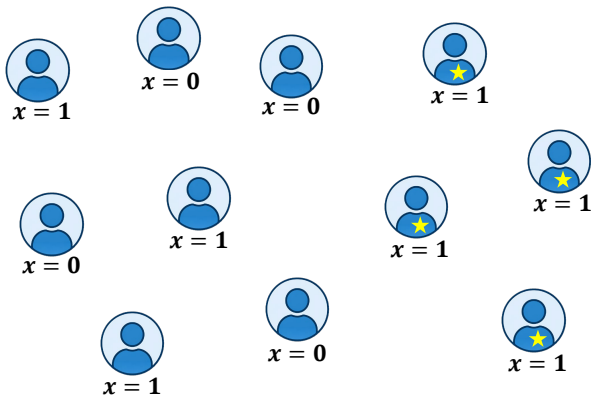
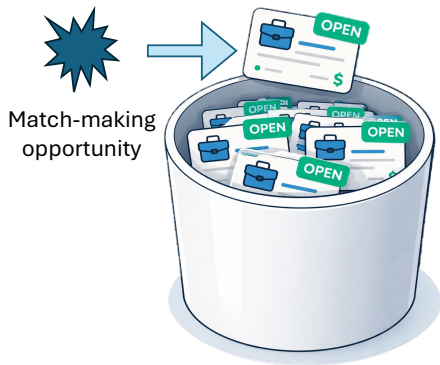


Match-making opportunities arrive at an exogenous Poisson rate λ .

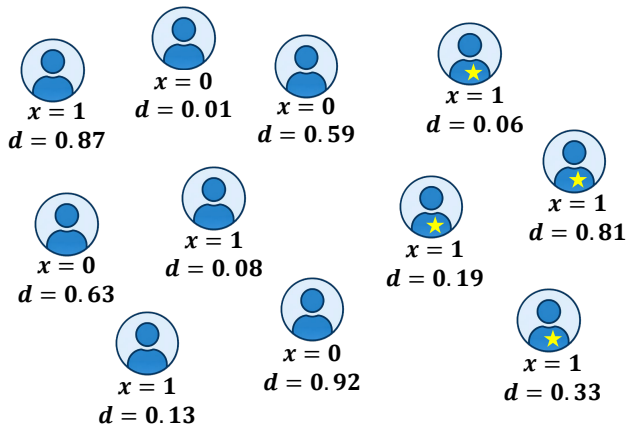
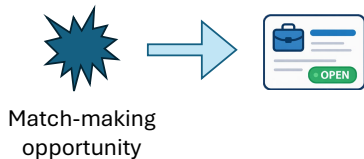

Match-making opportunity



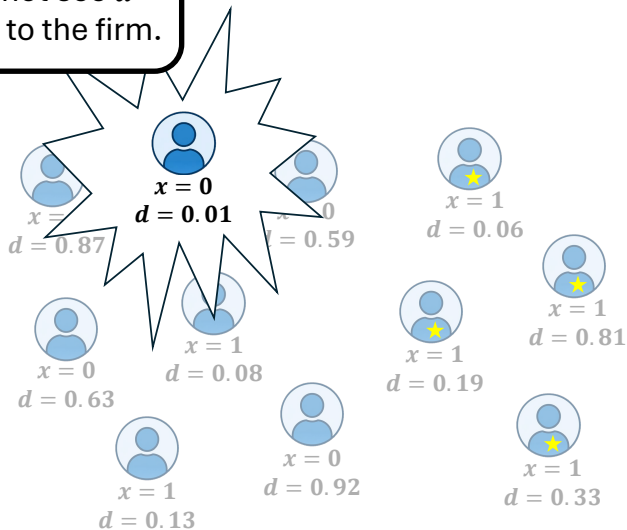
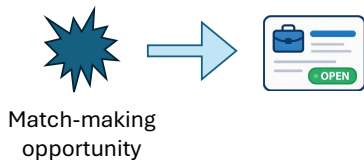
A random vacancy is drawn.



All match distances
are drawn.

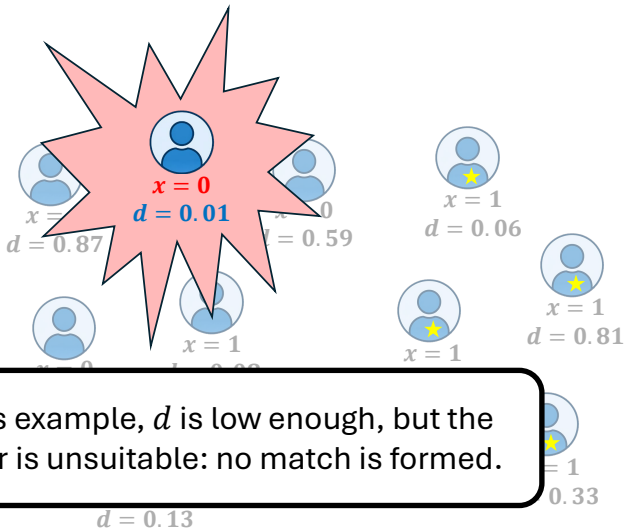


The platform identifies the lowest- d unemployed worker (but does not see d directly) and *recommends* them to the firm.

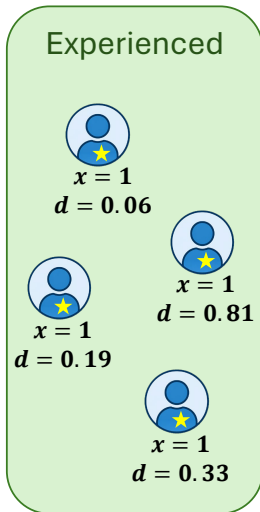
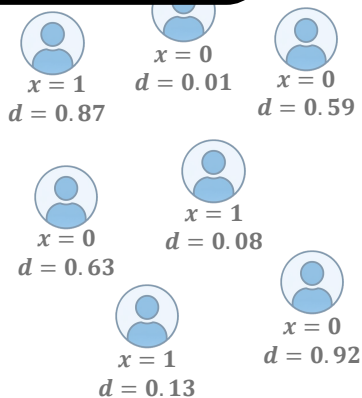




Match-making
opportunity



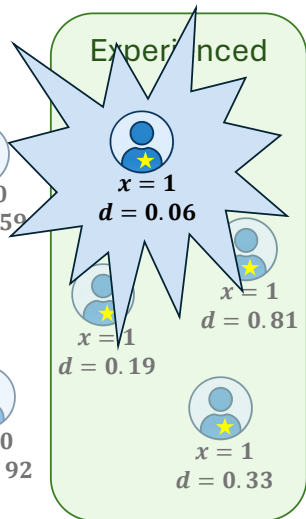
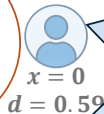
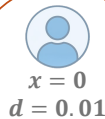
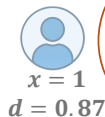
The platform may implement **steering**:
when a match-making opportunity arrives,
the *recommended worker* is the lowest- d
worker in the *experienced* pool.





Match-making
opportunity

Steering can sometimes **increase**
match formation because the pool
of experienced workers is
positively selected on suitability.



$d = 0.13$

In other cases, steering can **decrease** match formation, as it's easier to find a low-enough d in the unrestricted worker pool.

